HT-A_LS Series_V1.2

1. Document Description

1.1. Purpose of Compilation

This document defines the specification of Dyness_OpenAPI, which is open to the outside world by Dyness, systematically elaborates on the interface functions, business processes, communication protocols, and data format requirements, provides unified, standardized, and standardized access guidelines for partners, and ensures that the docking process is secure, efficient, and capable of sustainable evolution.

1.2. Platform Capability Description

Dyness_OpenAPI focuses on the full-scenario interaction requirements of household energy storage devices and opens the following capabilities to partners:

- Energy Storage Device Information Query
- Device Operating Status and Real-time Data Acquisition
- Query of Electricity Statistical Data
- Fault Information Acquisition
- Remote Control Parameter Query
- Control Instruction Issuance

After partners complete platform registration, certification, and authorization, they can interact with the energy storage system throughout the entire process based on standard APIs.

1.3. Version History

The interface version is iterated using a backward-compatible strategy. In case of incompatible updates, partners will be notified in advance to ensure the stability of the interface connection.

Version	Date	Description	Reviser
v1.0	2026/1/15	First Release	Product Manager.C
v1.2	2026/4/10	1. Add 6 types of control interfaces (Safety Regulation	Product Manager.C

		Country, Working Mode, Battery Parameter Settings, Load Control Settings, Peak Control Settings, Advanced Parameter Settings). 2. Delete the Control Query Service and Control Delivery Service interfaces. 3. Add access domain names (Global Site, North America, Latin America, Middle East, Africa, South America). 4. Add an example code module for authentication.	

2. System Architecture and Communication Specification

2.1. Communication Protocol

- Protocol Type: HTTPS
- Transport Encryption: SSL/TLS
- Data Format: JSON
- Character Encoding: UTF-7
- Default timeout: 30 seconds
- Recommended retry interval: 3 - 5 seconds

2.2. Access Domain Name

Select the corresponding domain name based on the region where the business belongs, and cross-region calls are not supported.

Region	Domain Name
Europe	https://eu-openapi.dyness.com/openapi/ems-device/

North America	https://na-openapi.dyness.com/openapi/ems-device/
Latin America	https://latam-openapi.dyness.com/openapi/ems-device/
Middle East	https://me-openapi.dyness.com/openapi/ems-device/
Africa	https://af-openapi.dyness.com/openapi/ems-device/
South America	https://sa-openapi.dyness.com/openapi/ems-device/
Global Site	https://global-openapi.dyness.com/openapi/ems-device/
China	https://openapi.dyness.com.cn/openapi/ems-device/
Asia-Pacific	https://apacopen-api.dyness.com/openapi/ems-device/

URL splicing rules: All interfaces use the POST method, and the request URL format is `<font style="color:rgb(15, 17, 21);background-color:rgb(235, 238, 242);">https://{Domain}/{Interface Path}</font>`, where `<font style="color:rgb(15, 17, 21);">background-color:rgb(235, 238, 242);">{Domain}</font>` Based on the regional selection (see the table above), `<font style="color:rgb(15, 17, 21);background-color:rgb(235, 238, 242);">{Interface Path}</font>` is the specific path defined for each interface.

2.3. Current Limiting Strategy

To ensure the stable operation of the system, interface calls must adhere to the following restrictions:

- Interface rate limit ⩽ 2 requests/second

If business requirements exceed the default limits, a written application may be submitted.

3. Security Authentication Mechanism

3.1. Authentication Method

The platform uses a self-signed authentication mechanism. All requests must carry the complete credentials in the Header.

Required Header Parameters:

** Header **	** Required **	** Instructions **
Authorization	Yes	Format: API {apiId}:{sign}
Content-MD5	Yes	Request Body MD5 Checksum
Content-Type	Yes	application/json

3.2. Example Code

代码块

```python
import json
import hashlib
import hmac
import base64
import time
from datetime import datetime, timezone
import urllib.request
import urllib.error
import sys

# --- CONFIGURATION ---
DEVICE_SN = "device sn"
APP_ID = "YOUR appId".strip()
APP_SECRET = "YOUR appSecret".strip()

# --- APAC SERVER ---
DOMAIN = "https://global-openapi.dyness.com"
PREFIX = "/openapi/ems-device"
INTERFACE = "/v2/GetRealTimeDataBySN"

def get_dyness_data():
    full_url = f"{DOMAIN}{PREFIX}{INTERFACE}"

    # 1. Body
    payload = {"sn": DEVICE_SN}
    # body_json = json.dumps(payload)
    # FIXME: Ensure the JSON is sorted and has no spaces for consistency
    body_json = json.dumps(payload, separators=(',', ':'), sort_keys=True)  # 与 JS 一致
```

```
29      body_bytes = body_json.encode('utf-8')
30      print(f"Body: {body_json}")
31      # 2. MD5
32      md5_hash = hashlib.md5(body_bytes).digest()
33      content_md5 = base64.b64encode(md5_hash).decode('utf-8')
34      print(f"Content-MD5: {content_md5}")
35
36      # 3. Date
37      now = datetime.now(timezone.utc)
38      weekdays = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
39      months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep",
    "Oct", "Nov", "Dec"]
40      date_str = f"{weekdays[now.weekday()]}, {now.day:02d} {months[now.month-
    1]} {now.year} {now.strftime('%H:%M:%S')} GMT"
41
42      # 4. Signature
43      # canonical_resource = f"{PREFIX}{INTERFACE}"
44      # FIXME: Remove PREFIX from canonical_resource
45      canonical_resource = f"{INTERFACE}"
46      string_to_sign =
    f"POST\n{content_md5}\napplication/json\n{date_str}\n{canonical_resource}"
47      print(f"String to Sign: {string_to_sign}")
48
49      secret_bytes = APP_SECRET.encode('utf-8')
50      sign_bytes = hmac.new(secret_bytes, string_to_sign.encode('utf-8'),
    hashlib.sha1).digest()
51      sign_base64 = base64.b64encode(sign_bytes).decode('utf-8')
52      print(f"Signature: {sign_base64}")
53
54      headers = {
55          "Content-Type": "application/json",
56          "Date": date_str,
57          "Content-MD5": content_md5,
58          "Authorization": f"API {APP_ID}:{sign_base64}",
59          "User-Agent": "HomeAssistant-Dyness"
60      }
61
62      try:
63          req = urllib.request.Request(full_url, data=body_bytes,
    headers=headers, method='POST')
64          with urllib.request.urlopen(req, timeout=30) as response:
65              # SUCCESS
66              print(response.read().decode('utf-8'))
67
68      except urllib.error.HTTPError as e:
69          # Handle "Waiting" state gracefully
70          if e.code == 401:
```

```
                print(json.dumps({
                    "code": "401",
                    "info": "Waiting for Key Activation...",
                    "data": {"soc": 0, "batteryPower": 0, "pvTotalPower": 0}
                }))
            elif e.code == 429:
                print(json.dumps({
                    "code": "429",
                    "info": "Connected (Rate Limit)",
                    "data": {"soc": 0, "batteryPower": 0, "pvTotalPower": 0}
                }))
            else:
                print(json.dumps({"code": str(e.code), "info": "HTTP Error",
"data": {}}))

        except Exception as e:
            print(json.dumps({"code": "500", "info": str(e), "data": {}}))

if __name__ == "__main__":
    get_dyness_data()
```

代码块

```
package com.dyness.ems;

import com.alibaba.fastjson.JSON;
import java.security.MessageDigest;
import java.security.NoSuchAlgorithmException;
import java.text.SimpleDateFormat;
import java.util.*;
import javax.crypto.Mac;
import javax.crypto.SecretKey;
import javax.crypto.spec.SecretKeySpec;
import okhttp3.MediaType;
import okhttp3.OkHttpClient;
import okhttp3.Response;
import org.apache.commons.codec.binary.Base64;

public class Authorization {

    public static void main(String[] args) {
        try {
            String key = "YOUR appId";
            String keySecret = "YOUR appSecret";
            Map<String, Object> map = new LinkedHashMap<>();
            map.put("pageNo", 1);
```

```
            map.put("pageSize", 10);
            String body = JSON.toJSONString(map);
            String ContentMd5 = getDigest(body);
            String Date = getGMTTime();
            String path = "/v1/device/getLastPowerDataBySn";
            String param =
                "POST" +
                "\n" +
                ContentMd5 +
                "\n" +
                "application/json" +
                "\n" +
                Date +
                "\n" +
                path;
            String sign = HmacSHA1Encrypt(param, keySecret);
            String url = "https://global-openapi.dyness.com/openapi/ems-device/" + path;
            OkHttpClient client = new OkHttpClient();
            MediaType xmlType = MediaType.parse(
                "application/json;charset=UTF-8"
            );
            okhttp3.RequestBody requestBody = okhttp3.RequestBody.create(
                xmlType,
                body
            );
            okhttp3.Request request = new okhttp3.Request.Builder()
                .url(url)
                .addHeader("Content-type", "application/json;charset=UTF-8")
                .addHeader("Authorization", "API " + key + ":" + sign)
                .addHeader("Content-MD5", ContentMd5)
                .addHeader("Date", Date)
                .post(requestBody)
                .build();
            Response response = client.newCall(request).execute();
            String string = response.body().string();
            System.out.println(string);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static String HmacSHA1Encrypt(String encryptText, String KeySecret)
        throws Exception {
        byte[] data = KeySecret.getBytes("UTF-8");
        // Constructs a key from the given byte array, the second parameter specifies the name of a key algorithm.
```

```
        SecretKey secretKey = new SecretKeySpec(data, "HmacSHA1");
        // Generate a Mac object that specifies the HmacSHA1 algorithm.
        Mac mac = Mac.getInstance("HmacSHA1");
        // Initialize the Mac object with the given key
        mac.init(secretKey);

        byte[] text = encryptText.getBytes("UTF-8");
        // Complete the encryption operation.
        byte[] result = mac.doFinal(text);
        return Base64.encodeBase64String(result);
    }

    public static String getGMTTime() {
        Calendar cd = Calendar.getInstance();
        SimpleDateFormat sdf = new SimpleDateFormat(
            "EEE, d MMM yyyy HH:mm:ss 'GMT'",
            Locale.US
        );
        // Setting the time zone to GMT
        sdf.setTimeZone(TimeZone.getTimeZone("GMT"));
        String str = sdf.format(cd.getTime());
        return str;
    }

    public static String getDigest(String test) {
        String result = "";
        try {
            MessageDigest md = MessageDigest.getInstance("MD5");
            md.update(test.getBytes());
            byte[] b = md.digest();
            result = java.util.Base64.getEncoder().encodeToString(b);
        } catch (NoSuchAlgorithmException e) {
            e.printStackTrace();
        }
        return result;
    }
}
```

4. Unified Response Specification

4.1. Response Format

All interfaces uniformly return a JSON structure, and each interface only retains the field table.

Field	**Type**	** Instructions **
code	String	Status Code
data	JSON	Business Data
info	String	Status Description

4.2. Error Code Description

Field	**Type**	** Instructions **
code	String	Status Code
data	JSON	Business Data
info	String	Status Description

5. Interface Classification Description

5.1. Classification of Business Information Exchange Functions

Business information exchange is divided into query and control types: only parameter tables are retained for each interface.

- **Query Class Interface:**
  - Power station information query;
  - Device information query;
  - Device basic information query;
  - Device status information query;
  - Device operation data query;
  - Device battery information query;
  - Device parallel operation data query;
  - Equipment fault information query;
- **Control Interface:**
  - Safety Regulations Country;
  - Working Mode;
  - Battery parameter settings;

- Load Control Settings:
- Peak control settings;
- Advanced Parameter Settings;

6. Query Interface Description

> This document is a user-side interface specification, retaining only the content required for invocation, and has removed internal code implementation descriptions, parameter details, and R&D-side remarks.

6.1. Device Information Query

6.1.1. Interface Definition

Project	Description
Interface Path	/v2/GetDeviceList

6.1.2. Input Parameters

Parameter Name	Type	Required or Not	Description
stationId	String	No	Unique identification code of the power station; when not passed, returns devices within the current permission scope

6.1.3. Input Parameter Example

代码块

```
{
  "stationId": "PLANT_001"
}
```

6.1.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
deviceSn	String	Yes	Device Serial Number
deviceModel	String	Yes	Device model, e.g., HT-A, LS series, etc.
workStatus	String	Yes	Working Status: `0: Standby`, `1: Grid-connected`, `2: Off-grid`, `3: Fault`, `4: Upgrade`, `5: Self-check`, `6: Generator Mode`
communicationStatus	String	Yes	Communication Status: `1: Online`, `2: Offline`, `3: Fault`

6.1.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": [
    {
      "deviceSn": "HT-0001",
      "deviceModel": "HT-A",
      "workStatus": "1",
      "communicationStatus": "1"
    },
    {
      "deviceSn": "HT-0001",
      "deviceModel": "LS",
      "workStatus": "0",
      "communicationStatus": "2"
    }
  ],
  "info": "success"
}
```

6.2. Device Basic Information Query

6.2.1. Interface Definition

Project	Description
Interface Path	/v2/GetDeviceInfBySN

6.2.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number

6.2.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001"
}
```

6.2.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
deviceSn	String	Yes	Device Serial Number
deviceModel	String	Yes	Device Model
firmwareVersion	String	No	Firmware Version (Version Number Concatenation)
inverterRatedPower	String	No	Inverter rated output power (unit: W)
batteryChargingRatedPower	String	No	Battery charging rated power, unit: W
batteryDischargingRatedPower	String	No	Battery discharge rated power, unit: W
acRatedOutputPower	String	No	

			AC rated output power, unit: W
acMaxOutputPower	String	No	AC maximum output power, unit: W
acMaxInputPower	String	No	AC maximum input power, unit W
acSinglePhaseMaxPower	String	No	AC single-phase maximum power, unit W
pvRatedPower	String	No	PV rated power, unit W
deviceType	String	No	Device Type: `0: Single-phase`, `1: Three-phase four-wire system`, `2: Three-phase three-wire system`, `3: Two-phase three-wire`

6.2.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "deviceSn": "HT-0001",
    "deviceModel": "HT-A",
    "firmwareVersion": "V2.3.1",
    "inverterRatedPower": "10",
    "batteryChargingRatedPower": "5000",
    "batteryDischargingRatedPower": "5000",
    "acRatedOutputPower": "10000",
    "acMaxOutputPower": "11000",
    "acMaxInputPower": "12000",
    "acSinglePhaseMaxPower": "3600",
    "pvRatedPower": "12000",
    "deviceType": "0"
  },
  "info": "success"
}
```

6.3. Device Status Information Query

6.3.1. Interface Definition

Project	Description
Interface Path	/v2/GetStatusInfBySN

6.3.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number

6.3.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001"
}
```

6.3.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
safeCountry	String	No	安规国家代码：`0:仓储/工厂测试`、`1:50Hz Grid default`、`2:60Hz Grid default`、`3:EN50549-1`、`4:EN50549-2`、`5:Italy CEI-021`、`6:Czech_A1`、`7:Czech_A2`、`8:Czech_B1`、`9:Germany(Hybrid) VDE-4105`、`10:Germany-B(AC

			coupling) VDE-4105、11:Austria、12:Poland、13:Denmark-West、14:Denmark-East、15:France、16:France Island 50Hz、17:France Island 60Hz、18:Australian-A、19:Australian-B、20:Australian-C、21:NewZealand、22:China、23:SouthAfrica、24:IEC61727、25:G98、26:G99、27:Spain-A、28:Spain-Island
runModel	String	No	EMS Operating Mode: 0: Invalid, 1: Self-use Mode, 2: ECO Mode, 3: Delay Charge Mode, 4: Fast Charge Mode, 5: Peak Shaving Mode, 6: Basic Mode (Battery Backup), 7: External Command Control, 8: Battery BMS Request Forced Charging, 9: Meter CT Self-check, 10: AU DRED0 Trigger, 11: Remote Shutdown, 12: Forced Off-grid Mode, 13: Forced Standby Mode (Allow Bypass), 14: Forced Stop Mode (Prohibit

			`Bypass)`, `15: Basic Mode (User Timer)`, `16: Ramadan Mode`
workStatus	String	No	PCS Operating Status: `0: Standby Status`, `1: Grid-connected Status`, `2: Off-grid Status`, `3: Fault Status`, `4: Firmware Upgrade Status`, `5: Self-test Mode`, `6: Generator Mode`
meterCommunication	String	No	Electric meter communication status: `0: Abnormal`, `1: Normal`, `2: Communication normal but CT abnormal`
antiReverseFlowStatus	String	No	Current anti-backflow status: `0: Off`, `1: On`
antiReverseFlowPower	String	No	Anti-reverse power Unit: W
pcsWorkStatus	String	No	PCS Operating Status: `0: Standby Status`, `1: Grid-connected Status`, `2: Off-grid Status`, `3: Fault Status`, `4: Firmware Upgrade Status`, `5: Self-test Mode`, `6: Generator Mode`

batteryCommunication	String	No	Battery communication status: `0: NG` , `1: OK`
gridConnectionStatus	String	No	Grid connection status: `0: Unknown` , `1: Normal` , `2: Recovering` , `3: Fault`
backupWorkStatus	String	No	Backup working status: `0: No output` , `1: Normal output`
generatorStatus	String	No	Generator Status: `0: Off` , `1: On`

6.3.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "safeCountry": "9",
    "runModel": "0x01",
    "workStatus": "1",
    "meterCommunication": "1",
    "antiReverseFlowStatus": "1",
    "antiReverseFlowPower": "1000",
    "pcsWorkStatus": "1",
    "batteryCommunication": "1",
    "gridConnectionStatus": "1",
    "backupWorkStatus": "1",
    "generatorStatus": "0"
  },
  "info": "success"
}
```

6.4. Equipment Operation Data Query

6.4.1. Interface Definition

Project	Description
Interface Path	/v2/GetRealTimeDataBySN

6.4.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	is	Device Serial Number

6.4.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001"
}
```

6.4.4. Return Fields

Return field name	Type	Is it mandatory to return?	Unit	Description
pvInfo.pvTotalPower	String	No	W	Total PV Power
pvInfo.thirdPartyInvPower	String	No	W	Third-party inverter power; some devices may be empty
gridInfo.gridStatus	String	No	-	Grid Status: `0: Unknown`, `1: Normal`, `2: Recovering`, `3: Fault`
gridInfo.activePower	String	No	W	Total active power of the power grid
gridInfo.reactivePower	String	No	Var	Total reactive power of the power grid

gridInfo.apparentPower	String	No	VA	Total apparent power of the power grid
gridInfo.gridVoltage	String	No	V	Grid Voltage
gridInfo.gridCurrent	String	No	A	Grid current
gridInfo.gridFrequency	String	No	Hz	Grid Frequency
gridInfo.totalPowerFactor	String	No	-	Total power factor of the power grid
gridInfo.inverterTotalPower	String	No	W	Total Inverter Power
meterInfo.meterType	String	No	-	Electric meter type code (`code:desc` , enumeration is subject to the device protocol)
meterInfo.meterCommunicationStatus	String	No	-	Electric meter communication status: `0: Abnormal` , `1: Normal` , `2: Communication normal but CT abnormal`
generatorInfo.generatorTotalPower	String	No	W	Total power of generator
generatorInfo.generatorVoltage	String	No	V	Generator Voltage
generatorInfo.generatorCurrent	String	No	A	Generator current
generatorInfo.generatorFrequency	String	No	Hz	Generator Frequency
	String	No	W	Total Load Power

loadInfo.loadTotalPower				
loadInfo.backupLoadPower	String	No	W	Backup Load Power
loadInfo.gridSideLoadTotalPower	String	No	W	Total power of grid-side load
loadInfo.backupPortLoadUtilization	String	No	%	Backup port load utilization
loadInfo.backupVoltage	String	No	V	Backup Voltage
loadInfo.backupCurrent	String	No	A	Backup current
loadInfo.backupFrequency	String	No	Hz	Backup Frequency
batteryInfo.batteryPower	String	No	W	Battery Power
batteryInfo.batteryVoltage	String	No	V	Battery Voltage
batteryInfo.batteryCurrent	String	No	A	Battery Current
batteryInfo.soc	String	No	%	Battery SOC
batteryInfo.soh	String	No	%	Battery SOH
batteryInfo.batteryStatus	String	No	-	Battery Charge and Discharge Status
batteryInfo.chargeCurrentLimit	String	No	A	Battery charging current limit
batteryInfo.dischargeCurrentLimit	String	No	A	Battery Discharge Current Limiting
batteryInfo.batteryTemperature	String	No	°C	Battery Temperature
batteryInfo.batteryCapacity	String	No	Ah	Battery Capacity

batteryInfo.batteryRatedCapacity	String	No	kWh	Battery Rated Capacity
batteryInfo.onGridDischargeDepth	String	No	%	Grid-connected discharge depth
batteryInfo.offGridDischargeDepth	String	No	%	Off-grid discharge depth
batteryInfo.cellMaxVoltage	String	No	V	Maximum voltage of a single cell
batteryInfo.cellMinVoltage	String	No	V	Cell Minimum Voltage
batteryInfo.cellMaxTemperature	String	No	°C	Maximum temperature of a single cell
batteryInfo.cellMinTemperature	String	No	°C	Cell Minimum Temperature
batteryInfo.bmsSoftwareVersion	String	No	-	BMS software version
batteryInfo.bmsHardwareVersion	String	No	-	BMS Hardware Version
batteryInfo.bmsCommunicationStatus	String	No	-	BMS communication status; may be empty when there is no upload

6.4.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "pvInfo": {
      "pvTotalPower": "5200",
      "thirdPartyInvPower": "0"
    },
    "gridInfo": {
      "gridStatus": "1",
      "activePower": "1500",
```

```
        "reactivePower": "0",
        "apparentPower": "1500",
        "gridVoltage": "230",
        "gridCurrent": "6.5",
        "gridFrequency": "50",
        "totalPowerFactor": "1",
        "inverterTotalPower": "4800"
      },
      "meterInfo": {
        "meterType": "1:Internal",
        "meterCommunicationStatus": "1"
      },
      "generatorInfo": {
        "generatorTotalPower": "0",
        "generatorVoltage": "0",
        "generatorCurrent": "0",
        "generatorFrequency": "0"
      },
      "loadInfo": {
        "loadTotalPower": "3600",
        "backupLoadPower": "800",
        "gridSideLoadTotalPower": "2800",
        "backupPortLoadUtilization": "0.2",
        "backupVoltage": "230",
        "backupCurrent": "3.5",
        "backupFrequency": "50"
      },
      "batteryInfo": {
        "batteryPower": "-1200",
        "batteryVoltage": "500",
        "batteryCurrent": "-2.4",
        "soc": "60",
        "soh": "98",
        "batteryStatus": "1",
        "chargeCurrentLimit": "80",
        "dischargeCurrentLimit": "80",
        "batteryTemperature": "28",
        "batteryCapacity": "280",
        "batteryRatedCapacity": "14.3",
        "onGridDischargeDepth": "20",
        "offGridDischargeDepth": "20",
        "cellMaxVoltage": "3.55",
        "cellMinVoltage": "3.45",
        "cellMaxTemperature": "30",
        "cellMinTemperature": "26",
        "bmsSoftwareVersion": "V1.2.0",
        "bmsHardwareVersion": "V1.0.0",
```

```
        "bmsCommunicationStatus": "1"
      }
    },
    "info": "success"
  }
```

6.5. Device Battery Information Query

6.5.1. Interface Definition

Project	Description
Interface Path	/v2/GetTotalEnergyDataBySN

6.5.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number

6.5.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001"
}
```

6.5.4. Return Fields

Return field name	Type	Is it mandatory to return?	Unit	Description
pvInfo.dailyPvGeneration	String	No	kWh	PV daily power generation
pvInfo.totalPvGeneration	String	No	kWh	PV Cumulative Power Generation
gridInfo.dailyBuyEnergy	String	No	kWh	Grid's daily electricity

				purchase volume
gridInfo.dailySellEnergy	String	No	kWh	Grid's daily electricity sales volume
gridInfo.totalBuyEnergy	String	No	kWh	Cumulative electricity purchased by the power grid
gridInfo.totalSellEnergy	String	No	kWh	Accumulated electricity sales of the power grid
generatorInfo.dailyGeneratorGeneration	String	No	kWh	Daily power generation of the generator
generatorInfo.totalGeneratorGeneration	String	No	kWh	Accumulated power generation of the generator
loadInfo.dailyElectricity	String	No	kWh	Load daily electricity consumption
loadInfo.totalElectricity	String	No	kWh	Load Cumulative Electricity Consumption
loadInfo.backupDailyEnergy	String	No	kWh	Backup daily electricity consumption
loadInfo.backupTotalEnergy	String	No	kWh	Backup Cumulative Electricity Consumption
loadInfo.gridLoadDailyEnergy	String	No	kWh	Daily electricity consumption of the grid-side load
loadInfo.gridLoadTotalEnergy	String	No	kWh	Cumulative electricity consumption of grid-side load

batteryInfo.dailyChargeEnergy	String	No	kWh	Daily charging capacity of the battery
batteryInfo.totalChargeEnergy	String	No	kWh	Cumulative battery charge
batteryInfo.dailyDischargeEnergy	String	No	kWh	Daily battery discharge
batteryInfo.totalDischargeEnergy	String	No	kWh	Cumulative battery discharge

6.5.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "pvInfo": {
      "dailyPvGeneration": "12.5",
      "totalPvGeneration": "3560.8"
    },
    "gridInfo": {
      "dailyBuyEnergy": "4.2",
      "dailySellEnergy": "1.1",
      "totalBuyEnergy": "780.3",
      "totalSellEnergy": "210.7"
    },
    "generatorInfo": {
      "dailyGeneratorGeneration": "0",
      "totalGeneratorGeneration": "12.4"
    },
    "loadInfo": {
      "dailyElectricity": "9.8",
      "totalElectricity": "2850.9",
      "backupDailyEnergy": "1.5",
      "backupTotalEnergy": "420.6",
      "gridLoadDailyEnergy": "8.3",
      "gridLoadTotalEnergy": "2430.3"
    },
    "batteryInfo": {
      "dailyChargeEnergy": "6.2",
      "totalChargeEnergy": "1980.4",
      "dailyDischargeEnergy": "5.7",
      "totalDischargeEnergy": "1760.9"
```

```
      }
    },
    "info": "success"
  }
```

6.6. Equipment Fault Information Query

6.6.1. Interface Definition

Project	Description
Interface Path	/v2/GetAlarmInfBySN

6.6.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
pageNum	Integer	No	Page number, default `1`
pageSize	Integer	No	Number of items per page, default `10`

6.6.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "pageNum": 1,
  "pageSize": 10
}
```

6.6.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
pageNum	Integer	Yes	Current Page Number

pageSize	Integer	Yes	Items per page
total	Long	Yes	Total number of records
list[].deviceSn	String	Yes	Device Serial Number
list[].eventGrade	String	Yes	Alarm Level
list[].eventType	Integer	Yes	Alarm Type
list[].eventCode	String	Yes	Event Code
list[].beginTime	String	Yes	Alarm time, format `yyyy-MM-dd HH:mm:ss`

6.6.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "pageNum": 1,
    "pageSize": 10,
    "total": 2,
    "list": [
      {
        "deviceSn": "HT-0001",
        "eventGrade": "2",
        "eventType": 1,
        "eventCode": "E1001",
        "beginTime": "2026-04-08 13:25:10"
      },
      {
        "deviceSn": "HT-0001",
        "eventGrade": "1",
        "eventType": 2,
        "eventCode": "W2003",
        "beginTime": "2026-04-08 12:40:05"
      }
    ]
  },
  "info": "success"
}
```

6.7. Parallel Machine Information Query

6.7.1. Interface Definition

Project	Description
Interface Path	/v2/GetParallelInfBySN

6.7.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number

6.7.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001"
}
```

6.7.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
parallelMyRule	String	Yes	Local machine parallel role: `0: Master`, `1: Slave`, `2: Other`, `3: New machine`
parallelNumber	String	Yes	Number of parallel units
deviceInfo[].deviceSn	String	Yes	Parallel device serial number
deviceInfo[].communicationStatus	String	Yes	Communication Status: `0: Abnormal`, `1: Normal`
deviceInfo[].deviceRole	String	Yes	

			Device Role: `0: Host`, `1: Slave`, `2: Other`, `3: New Machine`
deviceInfo[].registerStatus	String	No	Registration status; may be empty when the device does not return
pvTotalPower	String	No	Total parallel PV power, unit: W
batteryTotalPower	String	No	Total power of parallel batteries, unit: W
inverterTotalPower	String	No	Total parallel inverter power, unit W
gridTotalPower	String	No	Total power of the parallel grid, unit: W
backupLoadPower	String	No	Total power of parallel Backup load, unit: W
gridSideLoadTotalPower	String	No	Total power of the parallel grid-side load, unit: W
generatorPortTotalPower	String	No	Total power of parallel generator ports, unit: W

6.7.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "parallelMyRule": "0",
    "parallelNumber": "2",
    "deviceInfo": [
      {
        "deviceSn": "HT-0001",
        "communicationStatus": "1",
        "deviceRole": "0",
        "registerStatus": "1"
      },
```

```
        {
          "deviceSn": "HT-0001",
          "communicationStatus": "1",
          "deviceRole": "1",
          "registerStatus": "1"
        }
      ],
      "pvTotalPower": "5200",
      "batteryTotalPower": "-1200",
      "inverterTotalPower": "4800",
      "gridTotalPower": "1500",
      "backupLoadPower": "800",
      "gridSideLoadTotalPower": "2800",
      "generatorPortTotalPower": "0"
    },
    "info": "success"
  }
```


7. Control Class Interface Description

> ThYessection only retains the user call instructions, and the description of the underlying implementation details has been removed.

7.1. Control Query Service

7.1.1. Interface Definition

Project	Description
Interface Path	/v2/GetCtrlDataBySN

7.1.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
startPoint	Integer	Yes	Query start position (integer)
endPoint	Integer	Yes	Query end position (inclusive)

7.1.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "startPoint": 35205,
  "endPoint": 35210
}
```

7.1.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
data	Object	Yes	Query result object, key=parameter identifier, value=`code:desc` string; {} when there is no result

Example of the response body:

代码块

```
{
  "code": "200",
  "data": {
    "25243": "0",
    "25244": "1",
    "25356": "2"
  },
  "info": "Operation Successful"
}
```

7.2. Control Distribution Service

7.2.1. Interface Definition

Project	Description

Interface Path	/v2/SetCtrlDataBySN

7.2.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
controlData	Map<Integer, String>	Yes	Universal control parameter mapping; value uses `code:desc` string, the system parses it according to `code`, `desc` is only for readable description. value=null or empty string will be filtered

7.2.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "controlData": {
    "35205": "1",
    "35208": "3000"
  }
}
```

7.2.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
data	Object	Yes	Issued result object, key=parameter identifier, value=true/false

Example of response body:

代码块

```
{
  "code": "200",
  "data": {
    "25243": true,
    "25244": true
  },
  "info": "Operation Successful"
}
```

7.3. Battery Parameter Setting Service

7.3.1. Interface Definition

Project	Description
Interface Path	/v2/SetBatterySetting

7.3.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
batteryCommunicationProtocolCode	String	No	Battery communication protocol code: `0: Auto scan to match (without manual setting)`, `1: Dyness HV`, `2: BYD HV`, `3: Pylon HV`, `100: LithiumLV`, `101: Dyness LV`, `102: Pylon LV`, `254: NoBattery`, `255: LeadAcid`
batteryCapacity	String	No	Battery capacity, unit Ah
batteryEnergy	String	No	

			Battery capacity, unit kWh
batteryMaxChargeVoltage	String	No	Maximum charging voltage of the battery, unit V, value range `[0,1000]`
batteryMaxChargeCurrent	String	No	Maximum charging current of the battery, unit A
batteryMaxDischargeCurrent	String	No	Maximum discharge current of the battery, unit A
batteryMaxOverloadDischargeCurrent	String	No	Maximum overload discharge current of the battery, unit A
batteryOnGridUnderVoltageProtect	String	No	Battery grid-connected undervoltage protection point, unit V, value range `[0,1000]`
onGridDischargeDod	String	No	Battery grid-connected SOC protection point, unit %, value range `[0,100]`
batteryChargeLimit	String	No	Upper limit of battery grid-connected charging SOC, unit %, value range `[0,100]`
batteryOffGridUnderVoltageProtect	String	No	Off-grid undervoltage protection point of the battery, unit V, value range `[0,1000]`
offGridDischargeDod	String	No	Battery Off-Grid SOC Protection Point, unit %, value range `[0,100]`
batteryOffGridChargeLimit	String	No	Upper limit of battery off-grid charging SOC,

			unit %, value range `[0,100]`
batteryActivationVoltage	String	No	Battery activation voltage, unit V, value range `[40,60]` (only for low-voltage batteries)
batteryParallelEnable	String	No	Battery parallel enable during parallel operation: `0: Disable` , `1: Enable`

7.3.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "batteryCommunicationProtocolCode": "1",
  "batteryCapacity": "280",
  "batteryEnergy": "143",
  "batteryMaxChargeVoltage": "560",
  "batteryMaxChargeCurrent": "100",
  "batteryMaxDischargeCurrent": "100",
  "batteryMaxOverloadDischargeCurrent": "120",
  "batteryOnGridUnderVoltageProtect": "460",
  "onGridDischargeDod": "20",
  "batteryChargeLimit": "100",
  "batteryOffGridUnderVoltageProtect": "450",
  "offGridDischargeDod": "20",
  "batteryOffGridChargeLimit": "100",
  "batteryActivationVoltage": "480",
  "batteryParallelEnable": "1"
}
```

7.3.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.3.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```

7.4. Basic Parameter Setting Service

7.4.1. Interface Definition

Project	Description
Interface Path	/v2/SetBaseSetting

7.4.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
safetyCountry	String	No	安规国家代码：`0:仓储/工厂测试`、`1:50Hz Grid default`、`2:60Hz Grid default`、`3:EN50549-1`、`4:EN50549-2`、`5:Italy CEI-021`、`6:Czech_A1`、`7:Czech_A2`、`8:Czech_B1`、`9:Germany(Hybrid) VDE-4105`、`10:Germany-B(AC coupling) VDE-4105`、`11:Austria`、`12:Poland`、

			`13:Denmark-West`、`14:Denmark-East`、`15:France`、`16:France Island 50Hz`、`17:France Island 60Hz`、`18:Australian-A`、`19:Australian-B`、`20:Australian-C`、`21:NewZealand`、`22:China`、`23:SouthAfrica`、`24:IEC61727`、`25:G98`、`26:G99`、`27:Spain-A`、`28:Spain-Island`
nightDisChargeSwitch	String	No	Night Discharge Switch: `0: Off`, `1: On`
backwashSwitch	String	No	Anti-reverse switch: `0: Off`, `1: On`
hardwareBackwashSwitch	String	No	Hardware anti-backflow switch: `0: Off`, `1: On`
backwashPowerUpperLimit	String	No	Anti-reverse power upper limit, unit: W
selfControlPreference	String	No	Grid-feeding power error bias/self-consumption control preference, unit W, value range `[-500, 500]`
batterySocProtectSwitch	String	No	Battery SOC Protection Switch: `0: Off`, `1: On`
lowStrongChargePurchasePower	String	No	Battery low-power strong charging allowed purchase power (Pn), unit ‰, value range `[0,1000]`

battery1StrongChargeSocStart	String	No	Starting point of Battery 1 active forced charging SOC, unit %, value range `[0,50]`
battery1StrongChargeSocEnd	String	No	Active forced charging SOC endpoint of Battery 1, unit %, value range `[0, 50]`
generatorSwitch	String	No	Generator Function Enable: `0: Off` , `1: On and Connected to GEN Port` , `2: On and Connected to Grid Port`
generatorStartSoc	String	No	Generator start SOC, unit %, value range `[0, 100]`
generatorStopSoc	String	No	Generator stop SOC, unit %, value range `[0,100]`
generatorRatedPower	String	No	Generator rated power, unit W
generatorPortIntelLoad	String	No	Generator Port Intelligent Load Switch: `0: Off` , `1: On`
intelLoadOffGridEnable	String	No	Intelligent Load Off-Grid Enable: `0: Off` , `1: On`
intelLoadParallelGridEnable	String	No	Intelligent Load Grid-Connection Hold Enable: `0: Off` , `1: On`
intelLoadOpenPvPower	String	No	PV power threshold for intelligent load activation, unit: W
intelLoadOpenSoc	String	No	

			Intelligent load activation SOC, unit %, value range `[0,100]`
intelLoadCloseSoc	String	No	Intelligent Load Shutdown SOC, unit %, value range `[0,100]`
parallelInstallLocation	String	No	Grid-connected unit installation location: `0: Generator port` , `1: Backup port`
microGridEnable	String	No	Microgrid Enable: `0: Off` , `1: On`
microGridOverFrequency	String	No	Microgrid over-frequency, unit Hz, value range `[45,65]`

7.4.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "safetyCountry": "9",
  "nightDisChargeSwitch": "1",
  "backwashSwitch": "1",
  "hardwareBackwashSwitch": "1",
  "backwashPowerUpperLimit": "3000",
  "selfControlPreference": "50",
  "batterySocProtectSwitch": "1",
  "lowStrongChargePurchasePower": "500",
  "battery1StrongChargeSocStart": "20",
  "battery1StrongChargeSocEnd": "40",
  "generatorSwitch": "1",
  "generatorStartSoc": "30",
  "generatorStopSoc": "80",
  "generatorRatedPower": "5000",
  "generatorPortIntelLoad": "0",
  "intelLoadOffGridEnable": "1",
  "intelLoadParallelGridEnable": "1",
  "intelLoadOpenPvPower": "2500",
  "intelLoadOpenSoc": "60",
  "intelLoadCloseSoc": "40",
```

```
  "parallelInstallLocation": "1",
  "microGridEnable": "0",
  "microGridOverFrequency": "50"
}
```

7.4.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.4.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```

7.5. Working Mode Setting Service

7.5.1. Interface Definition

Project	Description
Interface Path	/v2/SetWorkModeSetting

7.5.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number

7.5.2.1. Basic Mode

Parameter Name	Type	Required or Not	Description
basicMode	Object	No	Base Mode Object
basicMode.basicWorkMode	String	No	Basic Mode Switch: `0: Off`, `0x01: Grid Stable`, `0x02: Grid Occasionally Outage`, `0x04: Grid Frequently Outage`, `0x08: Custom Mode 1`, `0x10: Custom Mode 2`
basicMode.occasionalOutage	Object	No	Occasionally stop electronic setting object
basicMode.occasionalOutage.chargePower	String	No	Charging power, unit W, value range `[0,999999]`
basicMode.occasionalOutage.chargeSoc	String	No	Charging SOC, unit %, value range `[0,100]`
basicMode.frequentOutage	Object	No	Frequent stop electronic setting object
basicMode.frequentOutage.chargePower	String	No	Charging power, unit W, value range `[0,999999]`
basicMode.frequentOutage.chargeSoc	String	No	Charging SOC, unit %, value range `[0,100]`
basicMode.customMode1Periods	Array	No	Custom Mode 1 Time Period
basicMode.customMode2Periods	Array	No	Custom Mode 2 Time Period

Object	Field	Type	Description
basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	periodNo	Integer	Time Period Number

basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	startTime	String	Start time, format `HH:mm`, example `08:30`
basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	endTime	String	End time, format `HH:mm`, example `18:00`
basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	targetSoc	String	Target SOC, unit %, value range `[0, 100]`
basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	powerLimit	String	Power limit, unit W, value range `[0,999999]`
basicMode.customMode1Periods[]/basicMode.customMode2Periods[]	gridChargeSwitch	String	Grid charging switch: `0: Off`, `1: On`

7.5.2.2. ECO Mode

Parameter Name	Type	Required or Not	Description
ecoMode	Object	No	ECO Mode Object
ecoMode.ecoModeSwitch	String	No	ECO Mode Switch: `0: Off`, `1: On`
ecoMode.ecoGroups	Array	No	ECO Group Configuration

Object	Field	Type	Description
ecoMode.ecoGroups[]	ecoGroup	Integer	ECO Group Number
ecoMode.ecoGroups[]	state	String	Group Switch: `0: Off`, `1: On`
ecoMode.ecoGroups[]	workStatus	String	Working Status: `0: Charging`, `1: Discharging`
ecoMode.ecoGroups[]	month	String	Month, supports CSV or bitmask; CSV example: `0,1,2` (0~11

			corresponds to January~December)
ecoMode.ecoGroups[]	week	String	Weekday, supports CSV or bitmask; CSV example: `0,1,2,3,4` (0 6 corresponds to Sunday and Saturday)
ecoMode.ecoGroups[]	startTime	String	Start time, format `HH:mm` , example `08:00`
ecoMode.ecoGroups[]	endTime	String	End time, format `HH:mm` , example `18:00`
ecoMode.ecoGroups[]	power	String	Power (Pn), unit ‰, value range `[0, 1000]`
ecoMode.ecoGroups[]	targetSoc	String	Target SOC, unit %, value range `[0,100]`

7.5.2.3. Delayed Charging Mode

Parameter Name	Type	Required or Not	Description
delayChargeMode	Object	No	Delayed Charging Mode Object
delayChargeMode.delayChargeSwitch	String	No	Delayed charging switch: `0: Off` , `1: On`
delayChargeMode.delayChargePeakPower	String	No	Peak power (Pn), unit ‰, value range `[0,1000]`
delayChargeMode.delayChargeStartTime	String	No	Start time, format `HH:mm` , example `01:00`
delayChargeMode.delayChargeEndTime	String	No	End time, format `HH:mm` , example `05:00`

7.5.2.4. Forced Off-Grid/Shutdown Mode

Parameter Name	Type	Required or Not	Description
forceOffGridModeSetting	Object	No	Forced Off-Grid/Shutdown Mode Object
forceOffGridModeSetting.forceOffGridMode	String	No	Mode: `0: Power Off`, `1: Forced Off-Grid`, `2: Standby Allowing Bypass`, `3: Standby Not Allowing Bypass`

7.5.2.5. PeakShaving模式

Parameter Name	Type	Required or Not	Description
peakShavingMode	Object	No	PeakShaving Mode Object
peakShavingMode.peakShavingSwitch	String	No	Switch: `0: Off`, `1: On`
peakShavingMode.peakShavingPowerLimit	String	No	Buying electricity limit power, unit W
peakShavingMode.peakShavingSocThreshold	String	No	SOC Threshold, unit %, value range `[0,100]`
peakShavingMode.peakShavingStartTime	String	No	Start time, format `HH:mm`, example `18:00`
peakShavingMode.peakShavingEndTime	String	No	End time, format `HH:mm`, example `22:00`

7.5.2.6. Ramadan Mode

Parameter Name	Type	Required or Not	Description
ramadanMode	Object	No	Ramadan Mode Object

ramadanMode.ramadanModeSwitch	String	No	Ramadan Mode Switch: `0: Off`, `1: On`
ramadanMode.ramadanPeriods	Array	No	Time Period Configuration

Object	Field	Type	Description
ramadanMode.ramadanPeriods[]	periodNo	Integer	Time Period Number
ramadanMode.ramadanPeriods[]	startTime	String	Start time, format `HH:mm`, example `03:00`
ramadanMode.ramadanPeriods[]	endTime	String	End time, format `HH:mm`, example `04:30`
ramadanMode.ramadanPeriods[]	targetSoc	String	Target SOC, unit %, value range `[0, 100]`
ramadanMode.ramadanPeriods[]	powerLimit	String	Power limit, unit W, value range `[0,999999]`
ramadanMode.ramadanPeriods[]	gridChargeSwitch	String	Grid charging switch: `0: Off`, `1: On`

7.5.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "basicMode": {
    "basicWorkMode": "1",
    "occasionalOutage": {
      "chargePower": "2000",
      "chargeSoc": "60"
    },
    "frequentOutage": {
      "chargePower": "2500",
      "chargeSoc": "70"
    },
```

```
    "customMode1Periods": [
      {
        "periodNo": 1,
        "startTime": "06:00",
        "endTime": "08:00",
        "targetSoc": "70",
        "powerLimit": "1800",
        "gridChargeSwitch": "1"
      }
    ],
    "customMode2Periods": [
      {
        "periodNo": 1,
        "startTime": "12:00",
        "endTime": "14:00",
        "targetSoc": "75",
        "powerLimit": "2000",
        "gridChargeSwitch": "0"
      }
    ]
  },
  "ecoMode": {
    "ecoModeSwitch": "1",
    "ecoGroups": [
      {
        "ecoGroup": 1,
        "state": "1",
        "workStatus": "1",
        "month": "0,1,2",
        "week": "0,1,2,3,4",
        "startTime": "08:00",
        "endTime": "18:00",
        "power": "2000",
        "targetSoc": "80"
      }
    ]
  },
  "delayChargeMode": {
    "delayChargeSwitch": "0",
    "delayChargePeakPower": "300",
    "delayChargeStartTime": "01:00",
    "delayChargeEndTime": "05:00"
  },
  "forceOffGridModeSetting": {
    "forceOffGridMode": "0"
  },
  "peakShavingMode": {
```

```
      "peakShavingSwitch": "1",
      "peakShavingPowerLimit": "3500",
      "peakShavingSocThreshold": "30",
      "peakShavingStartTime": "18:00",
      "peakShavingEndTime": "22:00"
    },
    "ramadanMode": {
      "ramadanModeSwitch": "0",
      "ramadanPeriods": [
        {
          "periodNo": 1,
          "startTime": "03:00",
          "endTime": "04:30",
          "targetSoc": "85",
          "powerLimit": "2200",
          "gridChargeSwitch": "1"
        }
      ]
    }
  }
```

7.5.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.5.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```

7.6. Advanced Parameter Setting Service

7.6.1. Interface Definition

Project	Description
Interface Path	/v2/SetAdvancedSetting

7.6.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
dredFunctionSwitch	String	No	DRED Function Switch: `0: Off`, `1: On`
rcrFunctionSwitch	String	No	RCR Function Switch: `0: Off`, `1: Normal RCR`, `2: Three-terminal Mode 14a EnWG`
turnOffFunctionSwitch	String	No	One-touch shutdown function switch: `0: Off`, `1: On`

7.6.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "dredFunctionSwitch": "1",
  "rcrFunctionSwitch": "1",
  "turnOffFunctionSwitch": "0"
}
```

7.6.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.6.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```

7.7. DSP Parameter Setting Service

7.7.1. Interface Definition

Project	Description
Interface Path	/v2/SetDspSetting

7.7.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
againGrid	String	No	Re-synchronization, value range `[0, 65535]` (common trigger value `122`)
againStart	String	No	Restart, value range `[0, 65535]` (common trigger value `123`)
pvStartGridVoltage	String	No	PV start-up grid-connected voltage, unit V, value range `[0,1000]`
isoDetectFaultThreshold	String	No	ISO insulation detection fault threshold, unit kΩ,

			value range `[0, 65000]`
isoInsulateDetectMode	String	No	ISO Insulation Detection Mode: `0: Detect for any grid connection`, `1: After power-on detection, no longer detect when switching from off-grid to on-grid`
pvShadowScanningSwitch	String	No	PV Shadow Scan Switch: `0: Off`, `1: On`
pvShadowScanningCycle	String	No	PV shadow scanning period, unit: min, value range `[0, 600]`
gridVoltageTraverseSwitch	String	No	Grid Voltage Ride-Through Switch: `0: Disable`, `1: Only LVRT`, `2: Only HVRT`, `3: LVRT & HVRT`
gridHighImpedanceMode	String	No	Grid High Resistance Mode: `0:Disable`, `1:Enable`
threePhaseUnbalanceOutputSwitch	String	No	Three-Phase Unbalance Output Switch: `0: Off`, `1: On`
threePhaseSequenceSwitch	String	No	Three-Phase Sequence Self-Adaptation Switch: `0: Off`, `1: On`
threePhaseFaultClear	String	No	Three-Phase Phase Sequence Fault Cleared
gridDciDetectionRatio	String	No	Grid-Connected DCI Detection Coefficient, Value Range `[0,20]`

backSwitch	String	No	Backup Switch: `0: Off`, `1: On`
backSwitchMode	String	No	Backup Switching Mode: `0: UPS Full Wave`, `1: UPS Half Wave`, `2: EPS`
offNetLowPowerChargeSwitch	String	No	Off-Grid Low-Power Charging Priority Switch: `0: Off`, `1: On`
offNetLowPowerChargeEndSoc	String	No	Off-grid Low Battery Priority Charging End SOC, Unit %
noBatteryPvOffNetworkSwitch	String	No	Battery-Free PV Off-Grid Switch: `0: Off`, `1: On`
offGridCounterReset	String	No	Off-grid overload counter reset, protocol range `[0,60000]` (commonly written as `1` to clear)
offGridTransferRelayTime	String	No	Off-grid to on-grid delay time, unit sec, value range `[0, 60000]`
offGridBackupUnderVoltageRatio	String	No	Off-grid backup undervoltage protection coefficient, unit %
offGridDcvProtectThreshold	String	No	Off-grid DCV protection threshold, unit V, value range [3, 100]
offGridPenRelaySwitch	String	No	Off-grid PE-N Relay Switch: `0: Off`, `1: On`

offGridPenRelayDetectSwitch	String	No	Off-grid PE-N Relay Detection Switch: `0: Off`, `1: On`
lightningProtectFailureAlarm	String	No	Lightning protection module failure alarm strategy: `0: No processing`, `1: Only prompt without shutdown`, `2: Alarm and shutdown`
powerPhaseShiftSwitch	String	No	Power Phase Shift Switch: `0: Off`, `1: On`
powerPhaseShiftTrigThreshold	String	No	Power phase-shift trigger threshold point, unit V, value range `[230, 300]`
gfciFaultManualClear	String	No	Manual Clear of GFCI Fault

7.7.3. Input Parameter Example

代码块

```
{
  "deviceSn": "HT-0001",
  "againGrid": "122",
  "againStart": "123",
  "pvStartGridVoltage": "500",
  "isoDetectFaultThreshold": "100",
  "isoInsulateDetectMode": "0",
  "pvShadowScanningSwitch": "1",
  "pvShadowScanningCycle": "10",
  "gridVoltageTraverseSwitch": "3",
  "gridHighImpedanceMode": "0",
  "threePhaseUnbalanceOutputSwitch": "1",
  "threePhaseSequenceSwitch": "1",
  "threePhaseFaultClear": "1",
  "gridDciDetectionRatio": "1",
  "backSwitch": "1",
  "backSwitchMode": "0",
```

```
    "offNetLowPowerChargeSwitch": "1",
    "offNetLowPowerChargeEndSoc": "35",
    "noBatteryPvOffNetworkSwitch": "0",
    "offGridCounterReset": "1",
    "offGridTransferRelayTime": "30",
    "offGridBackupUnderVoltageRatio": "60",
    "offGridDcvProtectThreshold": "50",
    "offGridPenRelaySwitch": "1",
    "offGridPenRelayDetectSwitch": "1",
    "lightningProtectFailureAlarm": "1",
    "powerPhaseShiftSwitch": "0",
    "powerPhaseShiftTrigThreshold": "230",
    "gfciFaultManualClear": "1"
  }
```

7.7.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.7.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```

7.8. ARM Parameter Setting Service

7.8.1. Interface Definition

Project	Description
Interface Path	/v2/SetArmSetting

7.8.2. Input Parameters

Parameter Name	Type	Required or Not	Description
deviceSn	String	Yes	Device Serial Number
extElectricMeter	String	No	External meter selection: `0: Built-in meter`, `1: External meter`
parallelHostCali	String	No	Parallel system master calibration: `0: Slave unit`, `1: Calibrated as system master`
pvGridInverterEnable	String	No	Monitoring PV grid-connected machine enable: `0: Do not monitor`, `1: External communication meter monitors grid-connected machine, system control automatically switches to internal meter`
Note: Setting extElectricMeter and pvGridInverterEnable simultaneously will cause a conflict. Please choose the setting according to the actual situation.			

7.8.3. Input Parameter Example

代码块

```
1  {
```

```
  "deviceSn": "HT-0001",
  "extElectricMeter": "1",
  "parallelHostCali": "1",
  "pvGridInverterEnable": "1"
}
```

7.8.4. Return Fields

Return field name	Type	Is it mandatory to return?	Description
1	Boolean	Yes	Unified Result Identifier

7.8.5. Return Value Example

代码块

```
{
  "code": "200",
  "data": {
    "1": true
  },
  "info": "Operation Successful"
}
```